

RAJ DESAI

Backend / Python Developer Intern

Ahmedabad, Gujarat | +91 8320283766 | rajgpdesai2007@gmail.com

github.com/RajDesai87 | linkedin.com/in/raj-desai132 | raj-desai-portfolio.vercel.app

SUMMARY

Computer Science Engineering student (3rd year) focused on backend development with Python, Django REST Framework, Flask, PostgreSQL, MySQL, Redis, and REST APIs. Experience building authentication systems (JWT, OAuth 2.0), asynchronous task processing, LLM/RAG-integrated systems, and modular multi-tier backend architectures. Strong foundation in Data Structures, OOP, and DBMS. Seeking a Backend / Python Developer internship.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL

Backend: Django, Django REST Framework, Flask, FastAPI, REST APIs, Celery

Databases: PostgreSQL, MySQL, SQLite, MongoDB, Redis

AI/ML: scikit-learn, XGBoost, FAISS, sentence-transformers, Groq/LLM APIs

Frontend: React, HTML, CSS, Tailwind CSS, Vite

Tools: Git, GitHub, Docker, Postman

Core: Data Structures, OOP, DBMS, API Design, Software Engineering

PROJECT EXPERIENCE

Mind Compass AI — AI-Powered Wellness Platform | [GitHub](#) | [Live Demo](#)

Django REST Framework, PostgreSQL, Redis, Celery, React, JWT, Google OAuth, Scikit-learn, NLTK/VADER

- Built a full-stack wellness platform (team project) with a Django REST Framework backend and React 19 + Vite frontend for mood tracking, journaling, analytics dashboards, and personalized recommendations.
- Implemented JWT authentication and Google OAuth 2.0, with Celery and Redis handling asynchronous task processing and background jobs.
- Integrated Scikit-learn, NLTK/VADER, and Joblib for sentiment-analysis and mood-prediction ML pipelines, plus Gemini and Groq APIs for conversational AI features.

Discussion-Den — Community Discussion Platform | [GitHub](#) | [Live Demo](#)

Python, Flask, SQLAlchemy, PostgreSQL, Flask-Login, Authlib, Flask-Limiter, Bootstrap 5, JavaScript

- Developed a Flask-based discussion platform (group project) with communities, posts, nested comments, voting, saved posts, and search.
- Implemented modular Blueprint routing and an application-factory structure following a 3-tier architecture.
- Added Google OAuth, session management, CSRF protection, rate limiting, and ORM-based database operations with SQLAlchemy, plus persona/identity-switching backed by PostgreSQL.

Stock Sentiment RAG & ML Prediction Dashboard | [GitHub](#)

Python, scikit-learn, XGBoost, FAISS, sentence-transformers, Groq LLM API, yfinance, Streamlit

- Built an end-to-end ML + RAG pipeline predicting next-day stock direction across 5 tickers by combining LLM-extracted news sentiment (Groq/Llama) with technical indicators, comparing Logistic Regression, Random Forest, and XGBoost with chronological train/test splitting to prevent temporal leakage.
- Identified and fixed a data-leakage bug where a synthetic sentiment feature was silently derived from same-day price returns for ~85% of rows, which had inflated reported accuracy by 14 points and Sharpe ratio to an artificial 3.66; re-validated results end-to-end and documented the audit in the README.
- Implemented a retrieval-augmented Q&A assistant using a FAISS vector index over embedded news articles, grounding Groq LLM responses in retrieved context and model predictions, with retry/backoff and rate-limit handling for API resilience.

EDUCATION

B.Tech in Computer Science Engineering — Lok Jagruti University, Ahmedabad, Gujarat

Sep 2024 – Sep 2028 | Currently in 5th Semester (3rd Year) | GPA: 9.07/10

CERTIFICATIONS

[Inheritance & Data Structures in Java](#) • [Intro to HTML, CSS & JavaScript](#) • [Intro to Java](#)
[Advanced Relational Database and SQL](#) • [Exploratory Data Analysis for Machine Learning](#)